



# PART 1: Beginner-Level Practical Tasks

## ● Task 1

Create a user named `developer` with:

- Home directory
- Bash shell

---

## ● Task 2

Set password for `developer`.

---

## ● Task 3

Force password change on next login.

---

## ● Task 4

Check password aging details of `developer`.

---

## ● Task 5

Lock and then unlock the user account.

---

## ● Task 6

Delete the user along with home directory.

---



## PART 2: Intermediate-Level Tasks

### ● Task 1 – Implement Company Password Policy

Set policy:

- Password expires in 60 days
- Minimum 7 days before change
- 10 days warning

Apply to user: `employee1`

---

### ● Task 2 – Create Temporary Employee

Create user `intern1`:

- Account expires on 31 Dec 2026
  - Force password change at first login
  - Default shell `/bin/bash`
- 

### ● Task 3 – Modify Default Settings

Change default settings so:

- All new users get `/bin/zsh`
- Password expires in 90 days
- Warning 7 days

Verify using:

`useradd -D`

---

## 🟡 Task 4 – Disable Inactive Accounts

Set account to disable 30 days after password expiry.

---

## 🟡 Task 5 – Enforce Strong Password Policy

Configure:

- Minimum 12 characters
  - At least 1 digit
  - At least 1 uppercase
  - At least 1 special character
- 

# PART 3: REAL-WORLD HANDS-ON PROJECT

## Enterprise Linux User Security Implementation

---

### Scenario

You are hired as a **Linux System Administrator** in a mid-size IT company.

The company recently failed a **security audit** because:

- Passwords never expire
- No password complexity enforcement

- Old employees still have active accounts
- No lockout policy for failed logins
- Temporary employees are not auto-expiring

Your task is to **secure and standardize user management across the server.**

---

## PROJECT OBJECTIVES

You must:

1. Implement company-wide password policy
  2. Configure password complexity rules
  3. Configure account lockout mechanism
  4. Create permanent and temporary users
  5. Configure account expiry
  6. Apply policy to existing users
  7. Verify everything like a real audit
- 

## Company Policy

### Password Policy

- Password expires every **60 days**
- Minimum 7 days before change
- Warning 10 days before expiry

- Lock account after 5 failed attempts
  - Unlock after 15 minutes
  - Prevent reuse of last 5 passwords
  - Minimum 12 characters
  - Must contain:
    - 1 uppercase
    - 1 lowercase
    - 1 digit
    - 1 special character
- 

## Employees to Create

| Username     | Role       | Account Type | Expiry      |
|--------------|------------|--------------|-------------|
| emp01        | Developer  | Permanent    | Never       |
| emp02        | Tester     | Permanent    | Never       |
| intern01     | Intern     | Temporary    | 31 Dec 2026 |
| contractor01 | Contractor | Contract     | 31 Dec 2026 |
| auditor01    | Auditor    | Temporary    | 31 Dec 2026 |

---

## Phase 7 – Security Testing

### Test 1: Password Complexity

Try weak password → Should fail.

### Test 2: Password Reuse

Change password 5 times → Try old one → Should fail.

### Test 3: Failed Login

Enter wrong password 5 times → Account should lock.

---

## Deliverables (If This Was a Real Assignment)

You must submit:

1. Screenshot of `/etc/login.defs`
2. Screenshot of `/etc/security/pwquality.conf`
3. PAM configuration screenshot
4. Output of `chage -l` for each user
5. Output of `useradd -D`
6. Demonstration of lockout test